First year Aerospace Engineering

Introduction to Aerospace Structures

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What is the main objective?

Less weight – less fuel consumption – lower emissions etc.

Because airlines want to save the planet?

- In part, yes (as that's what customers and legislators want)
- But mainly: save on fuel **cost**

Lower weight leads to cheaper operational costs

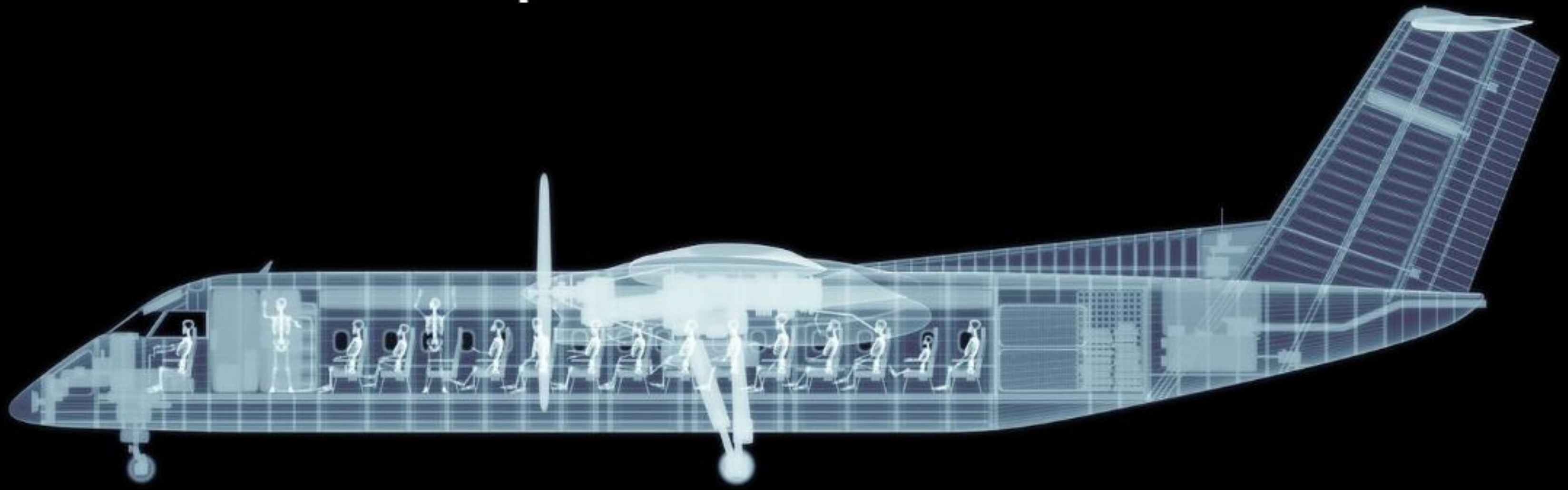
Role of Weight

Aircraft (engines) burn fuel to generate **Thermal Energy**
which is converted to **Kinetic Energy**
which is transferred to motion
using **Work** in the form of **Thrust** (T), times **Displacement** (d)

$$\left.\begin{aligned} Energy &= Work = T \cdot d \\ T &= \frac{C_D}{C_L} \cdot W \end{aligned}\right\} \quad Energy \cong \frac{C_D}{C_L} \cdot W \cdot d$$

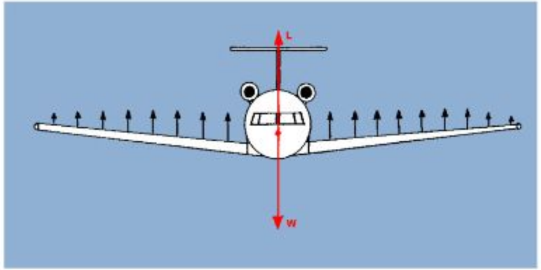
What is an aircraft structure?

The “stressed” part of the aircraft



Functions of a structure:

Carry loads



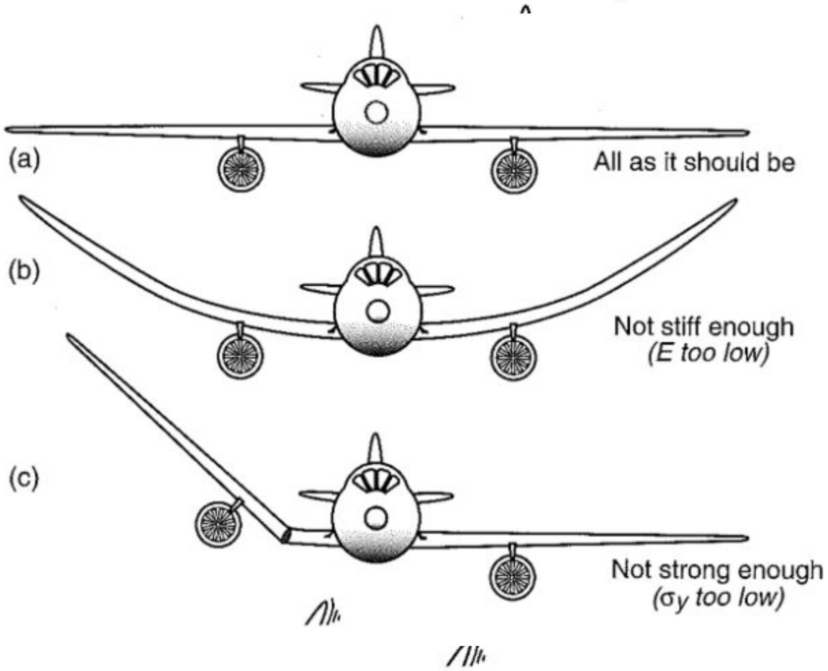
Protect passengers and cargo

Maintain aerodynamic shape

Connect subsystems

Requirements of an aircraft structure

- Strength
- Stiffness
- Lightweight
- Durable
- Cost-effective
- Maintainable



What is an airframe

An **airframe** is the basic structure of an aircraft or spacecraft, which does not contain any equipment or interior installations.

Primary and secondary structures

- A **primary structure** or **Principal Structural Element** (PSE) is a critical load-bearing structure of an aircraft that, in the case of severe damage, will cause the failure of the entire aircraft.
    - **Failure is catastrophic**

- A **secondary structure** or **Non-Principal Structural Element** is a structural element of an aircraft or spacecraft that carries only aerodynamic and inertial loads generated on or in the structure
    - **Failure is NOT catastrophic**

Dorsal Fin

Loading

5 Types of loads transmitted by structures

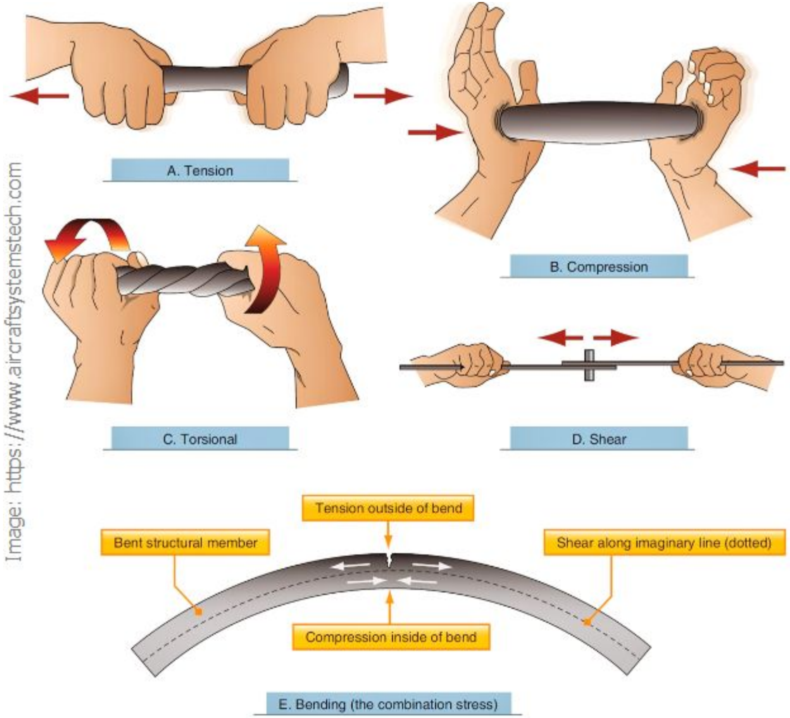
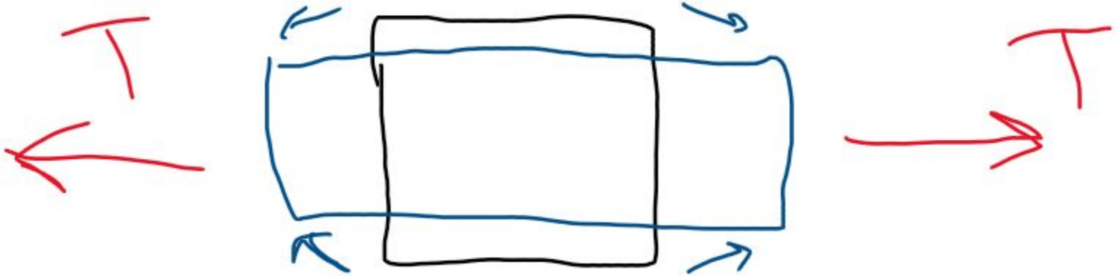


Image: https://www.aircraftsystemstech.com

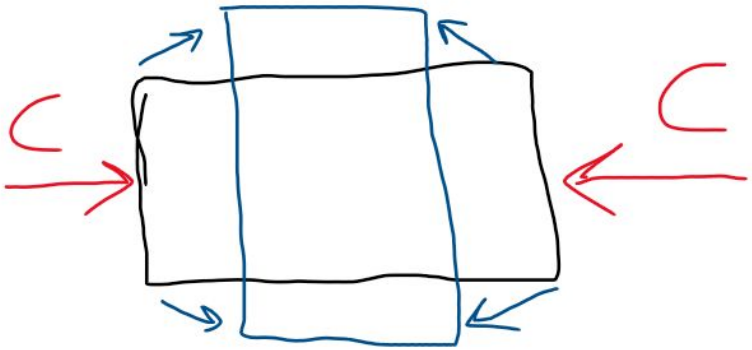
Tension

A force pulling an object apart



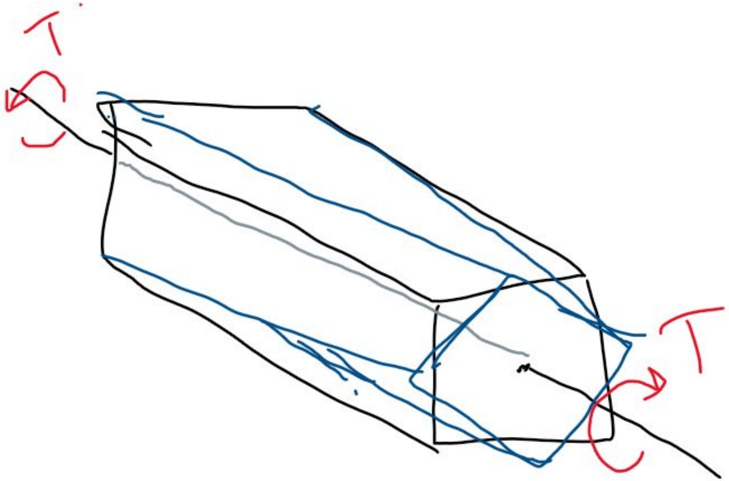
Compression

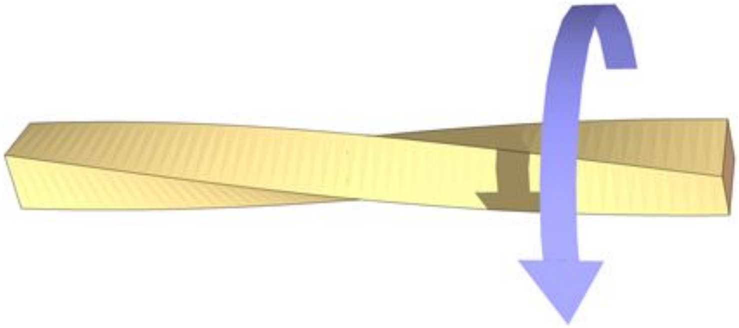
A force pushing an object together



Torsion

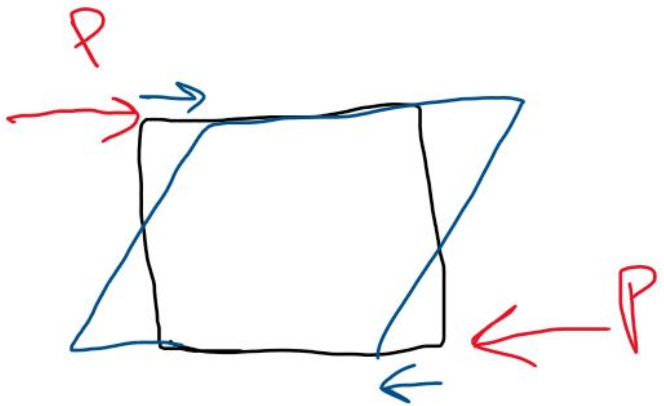
The action of twisting
Moment (torque) loading along longitudinal axis





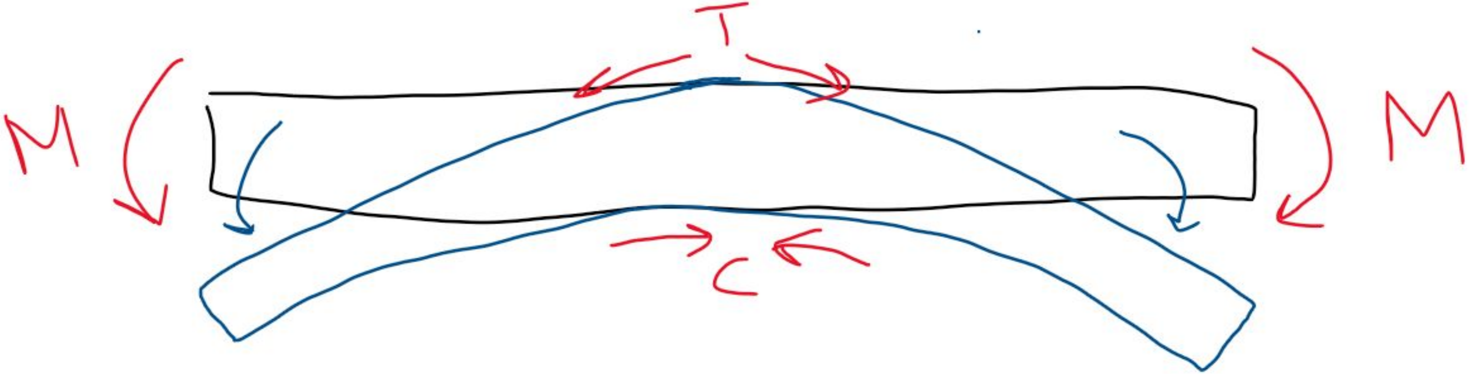
Shear

Force that acts parallel or tangential to a surface, causing it to deform or slide.
Pair of equal forces acting in opposite directions along the two faces of the layer



Bending

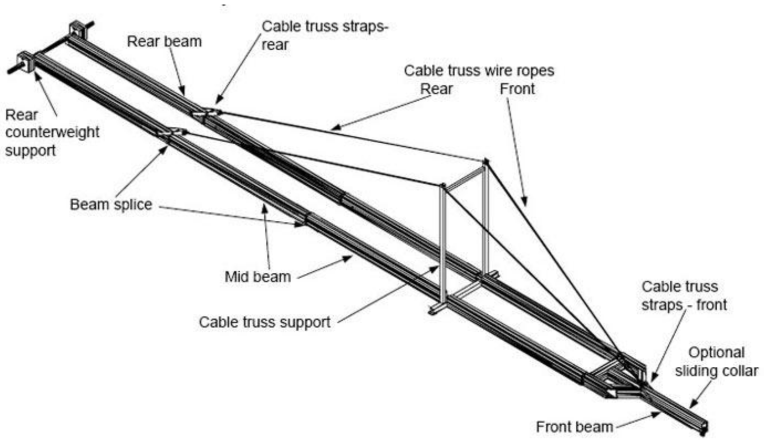
Caused by an external force or a moment, that causes a structure to be
Bending is a combination of tension, compression and shear



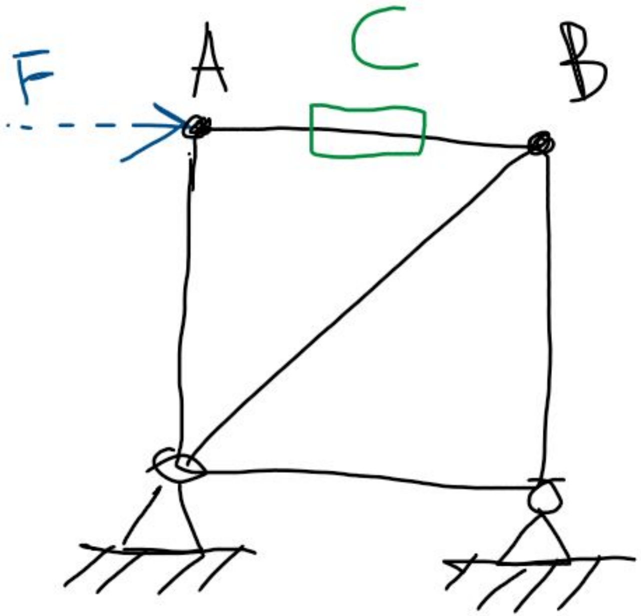
Structural concepts

Truss structure

Truss structures are made of bars, tubes and wires, which carry all loads. Structural rigidity is obtained with the diagonal function, i.e. wires or rods placed under 45°.



Solving loads



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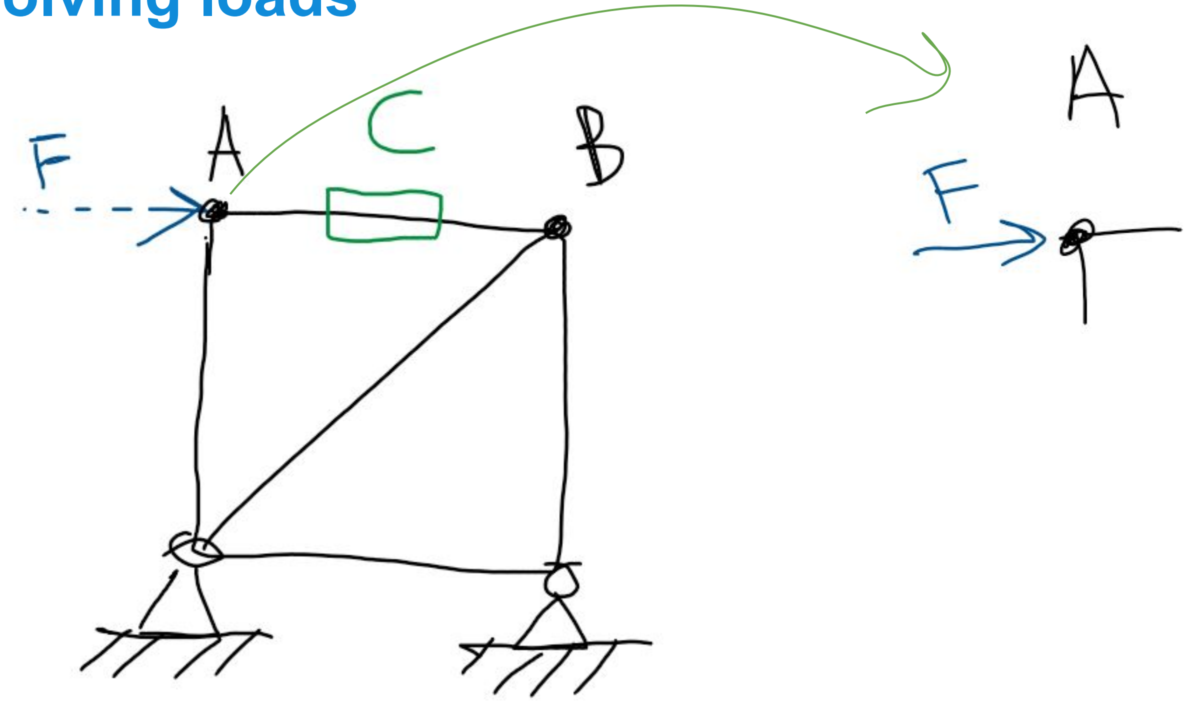

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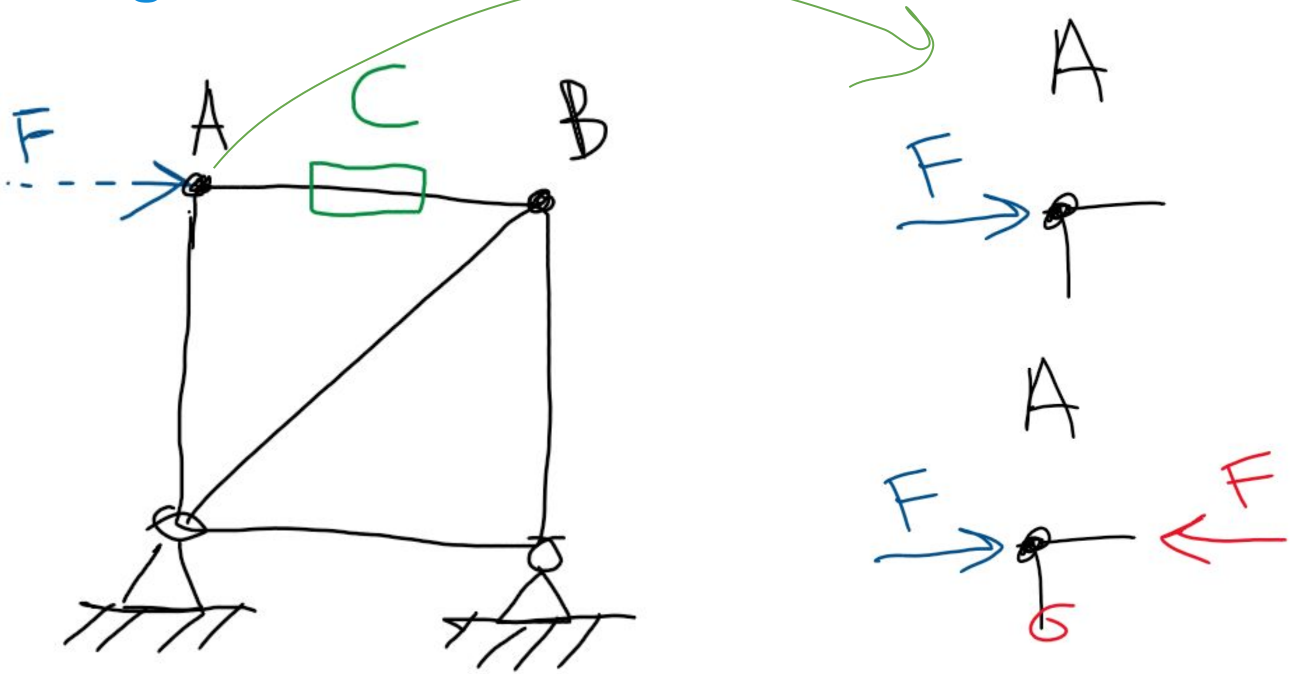
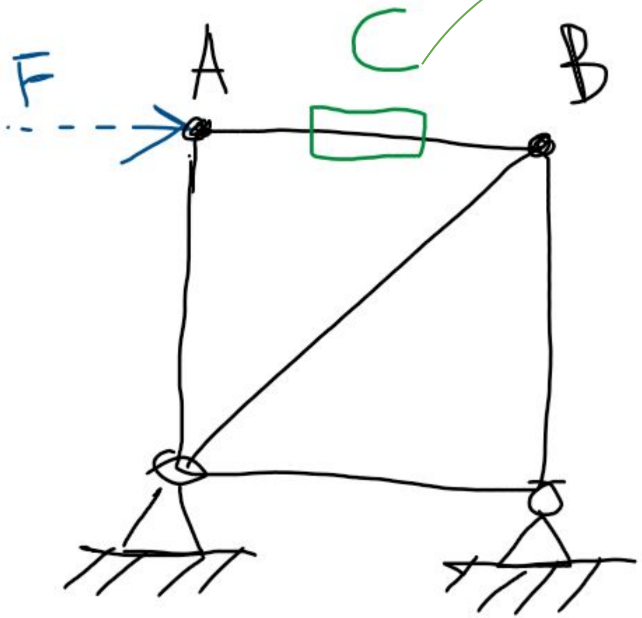
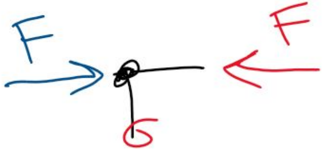
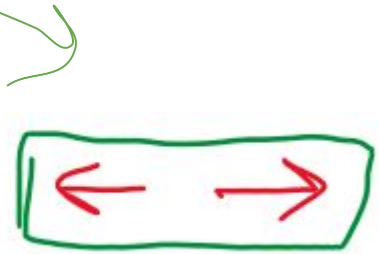


Rod is pushing out

Solving loads



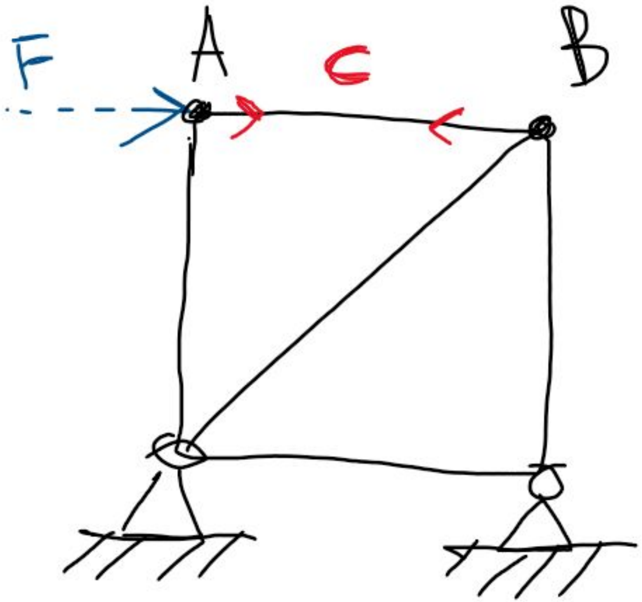


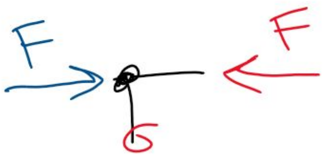


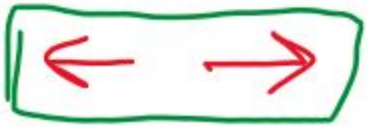
Rod is pushing out

Rod is under compression

Solving loads







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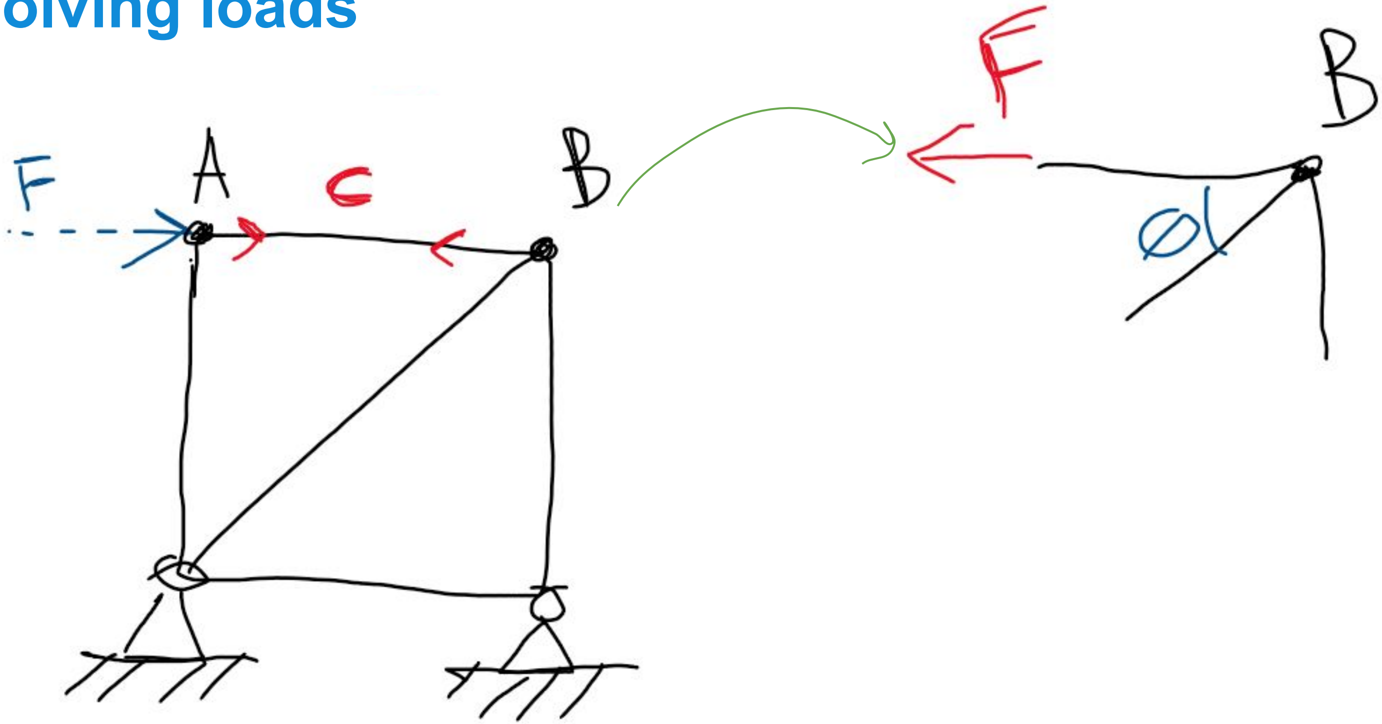

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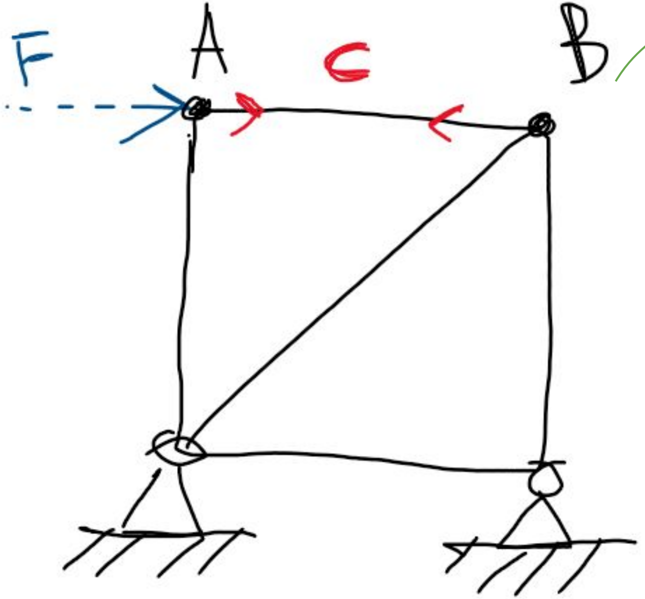

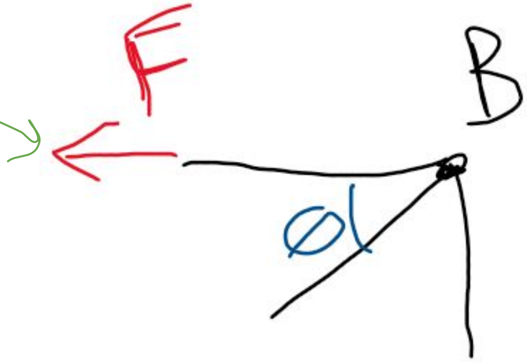

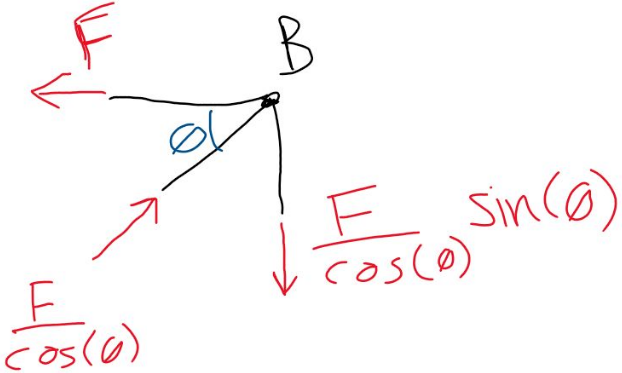

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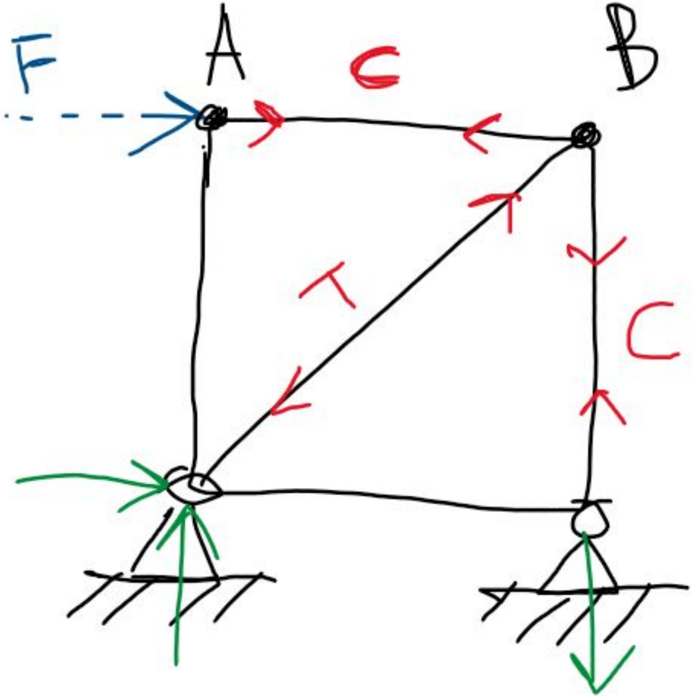
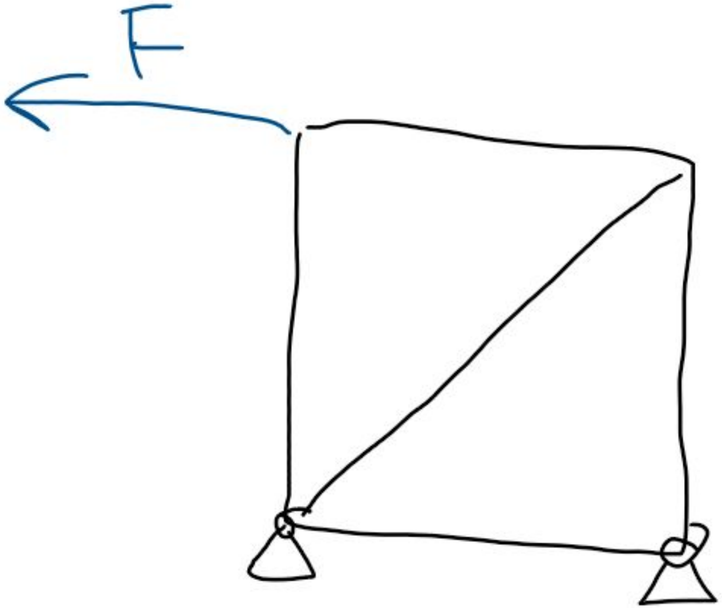


Solving loads

The problem of wires:



Solving loads

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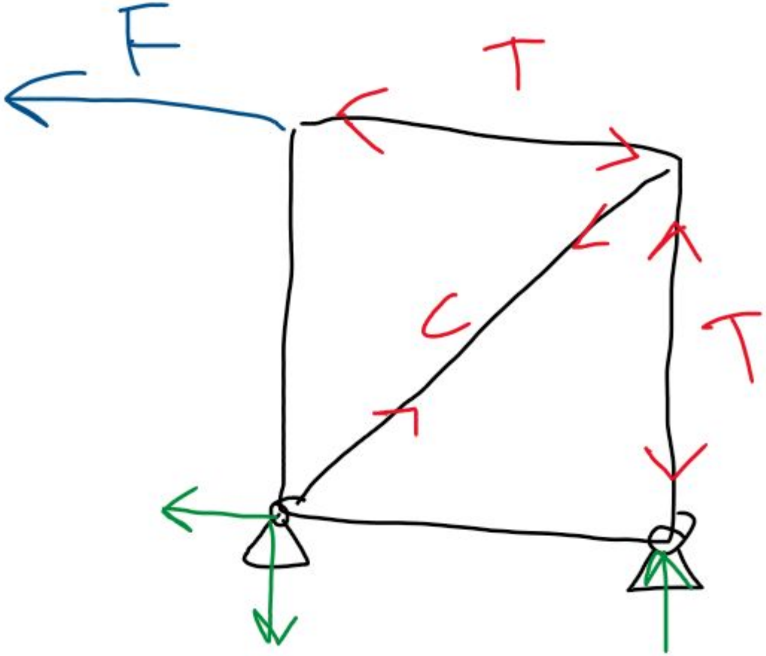

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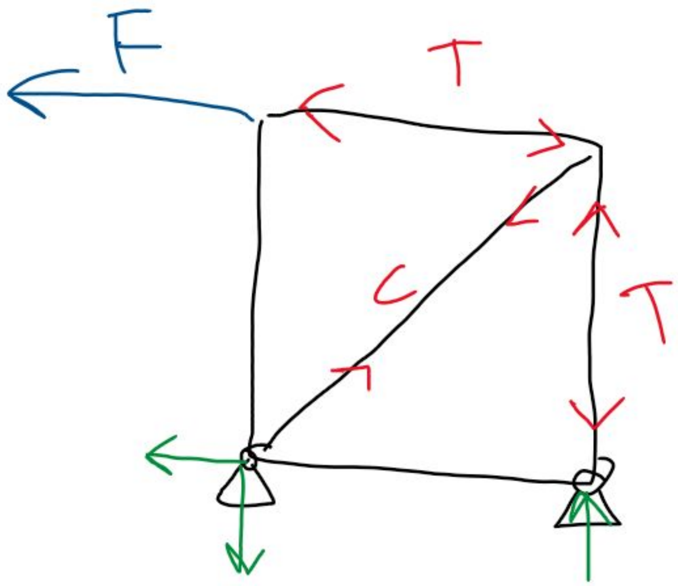
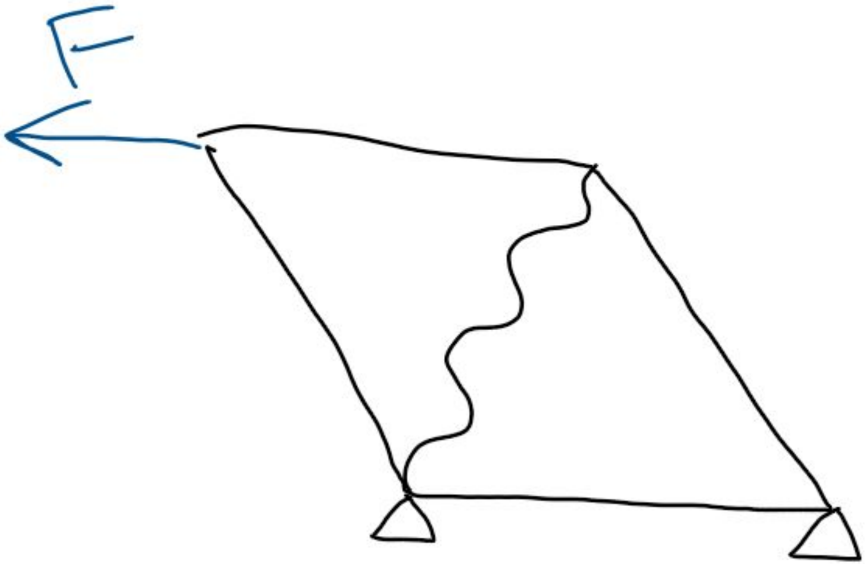


Wires cannot withstand compression!

Solving loads



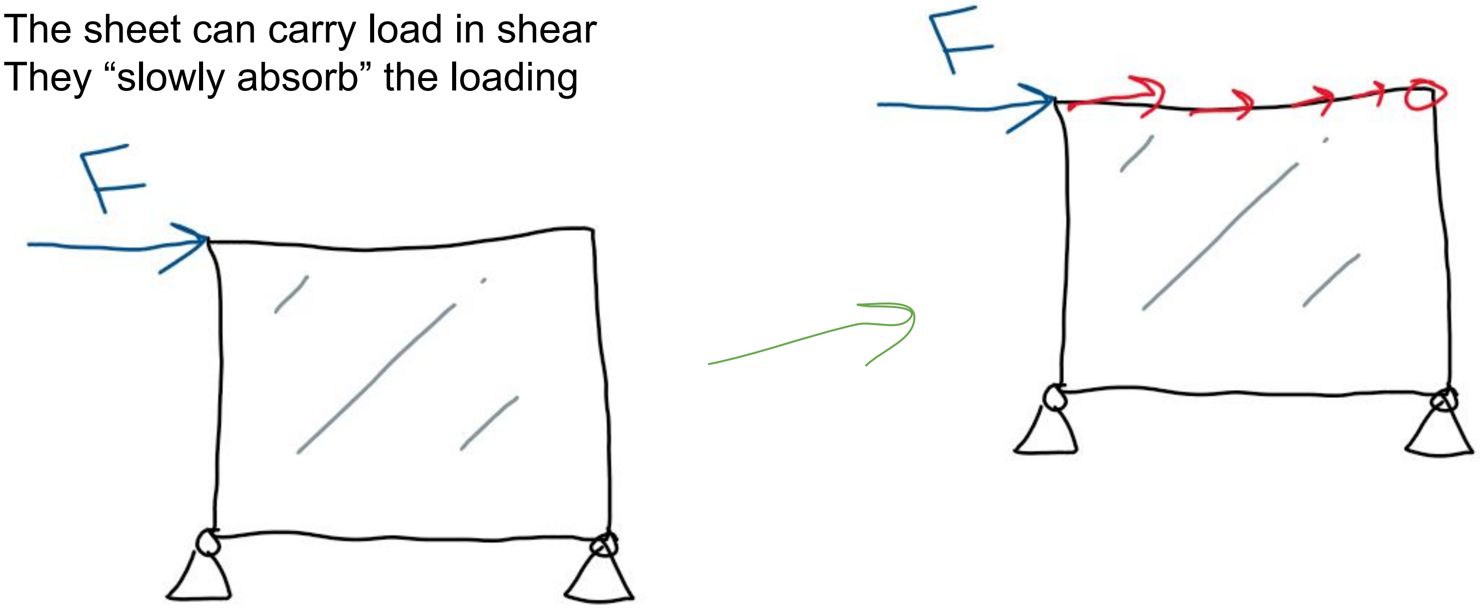
Wires cannot withstand compression! Structure fails

Shell structure

This concept relates to the application of sheet material in a truss structure to fulfil the diagonal function.
In addition to tension and compression (also with tubes and wires), sheet material can carry load in shear (diagonal function), but can also seal a structure from its environment

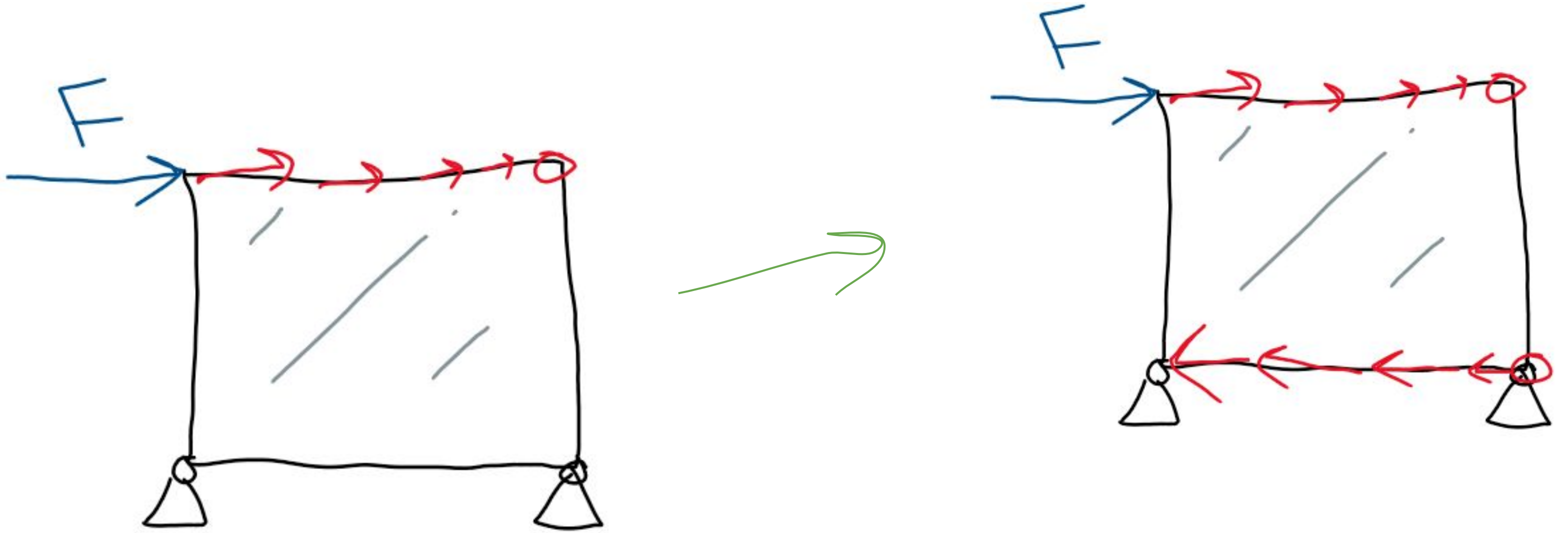
Shell structures

The sheet can carry load in shear
They “slowly absorb” the loading



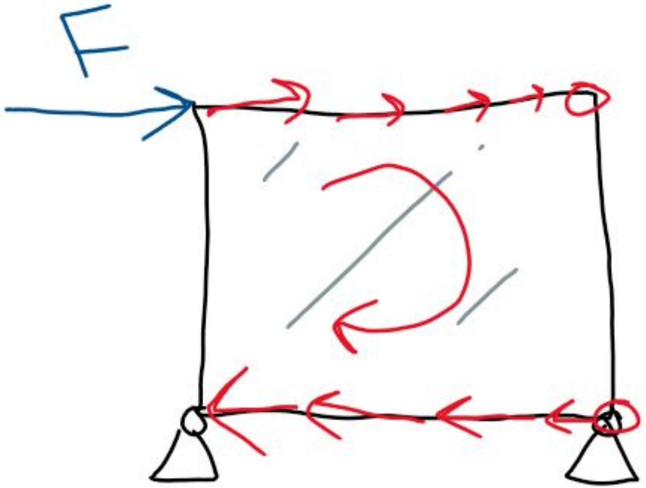
Shell structures

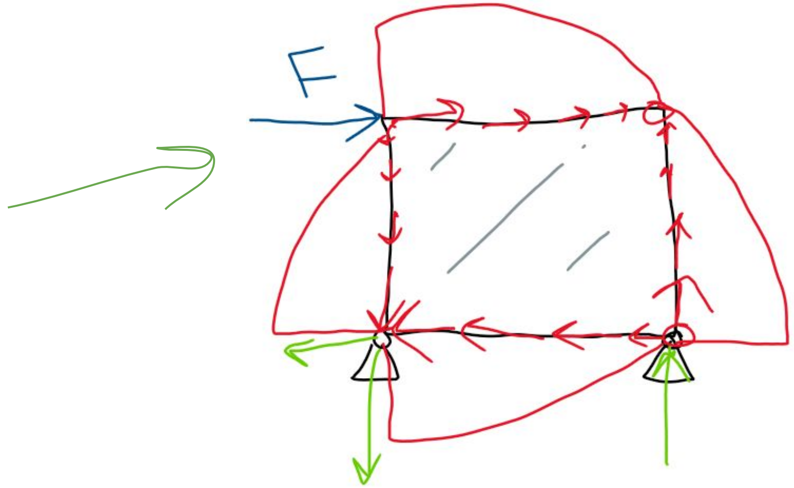
Remember shear **always** works in pairs



Shell structures

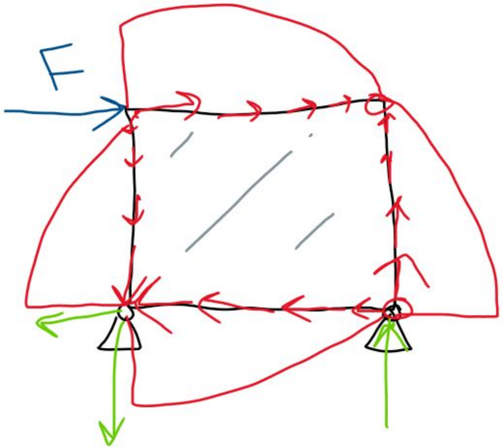
Remember shear **always** works in pairs
This would cause a moment, so there has to be vertical shear too

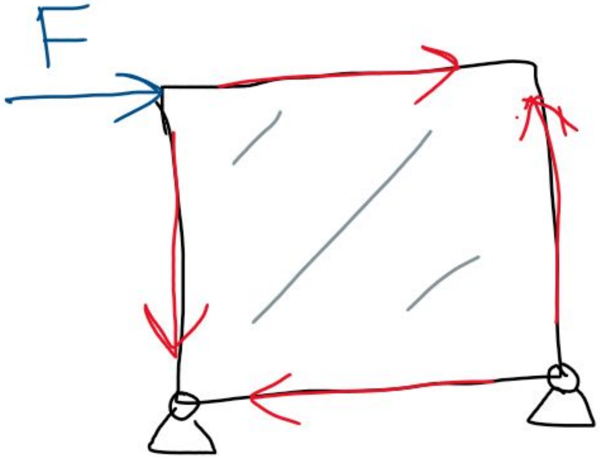


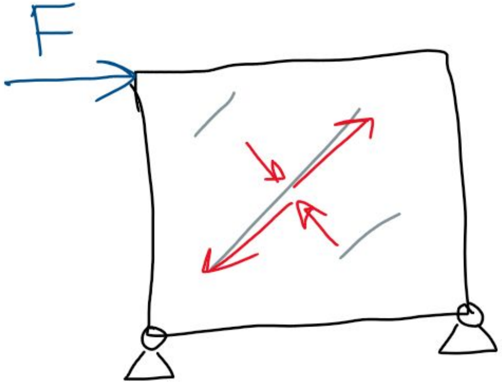


Shell structures

Simpler ways to draw shear:

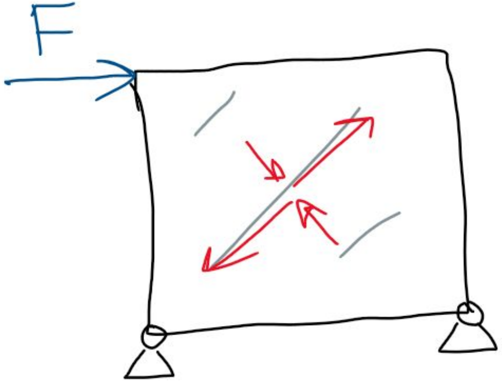






Shell structures

What happens if the stress is too large?



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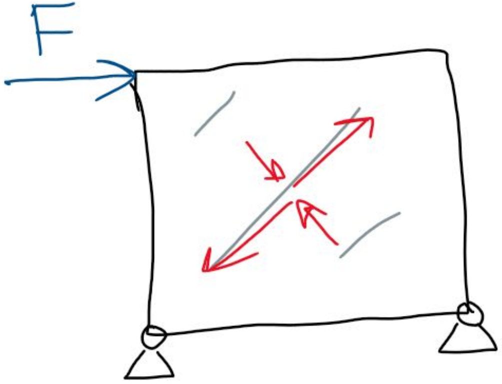

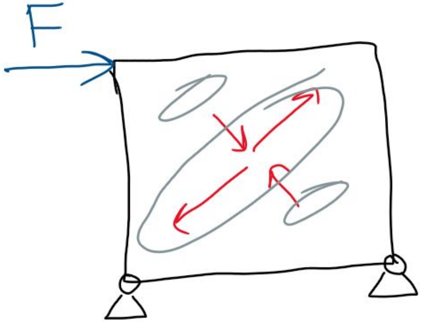

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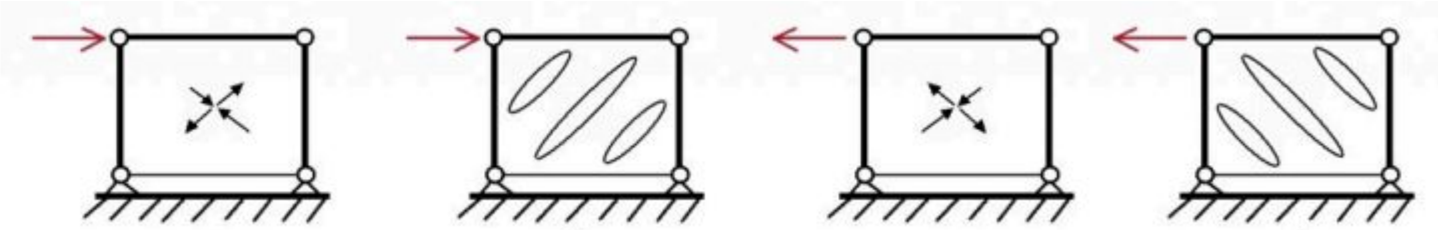

Shell structures

Although shell structures can withstand tension quite well, they cannot withstand compression well

Try with a piece of paper

Shell structures

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Try with a piece of paper

Reinforcements are needed

Reinforcements (stiffeners)

Stiffness is not only a material property but also a geometry parameter
Try geometrically stiffening your piece of paper

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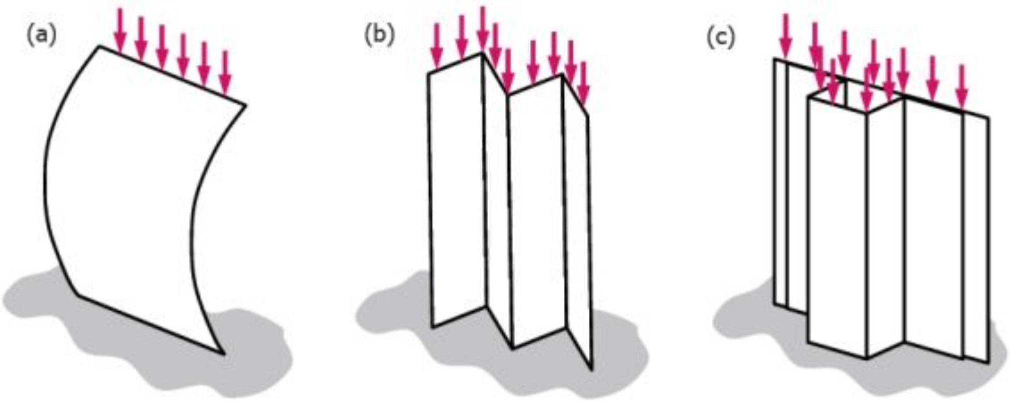
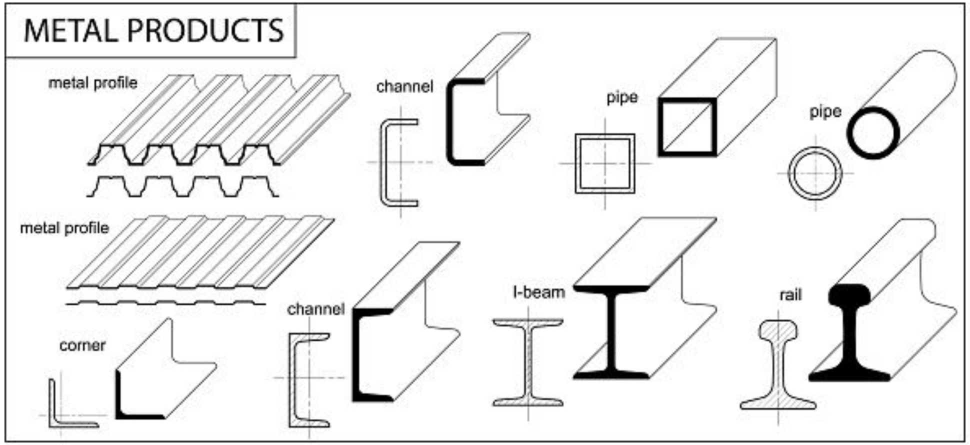


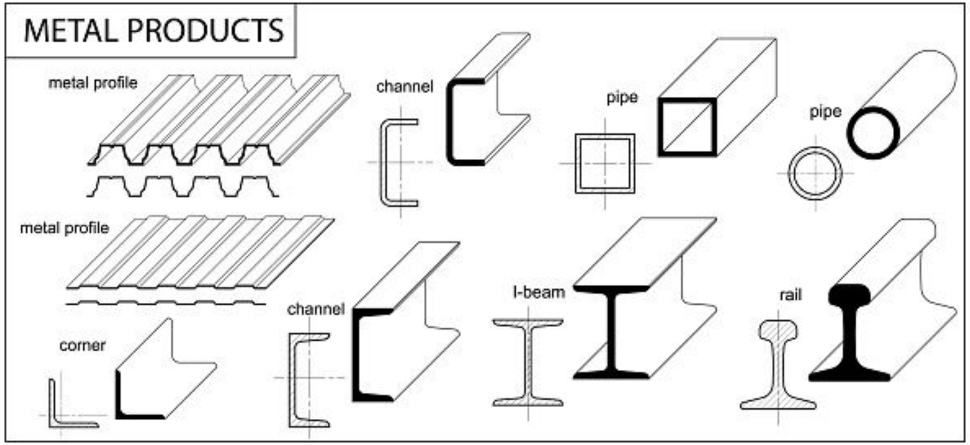
Reinforcements (stiffeners)

You can add stiffeners to improve the compression behaviour of sheets
There are various types of stiffeners



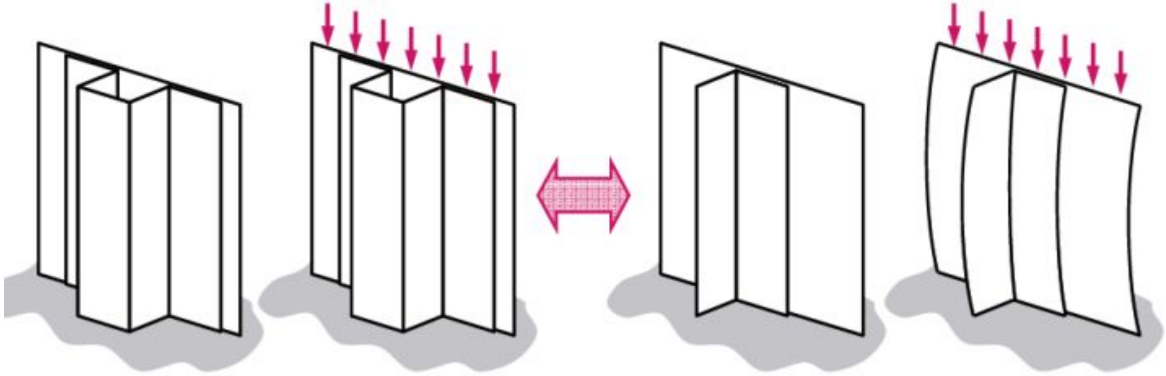
Reinforcements (stiffeners)

You can add stiffeners to improve the compression behaviour of sheets
There are various types of stiffeners
The more material further away from the sheet, the stiffer it is



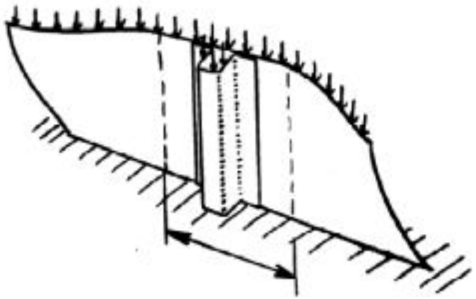
Reinforcements (stiffeners)

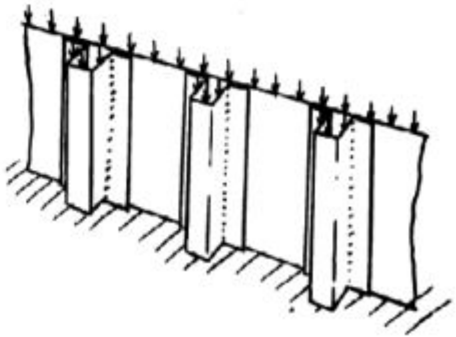
You can add stiffeners to improve the compression behaviour of sheets
There are various types of stiffeners
The more material further away from the sheet, the stiffer it is
So hat-stiffeners are more stable than L-stiffeners for example (but more expensive and complex, and don't allow for inspection)



Reinforcements (stiffeners)

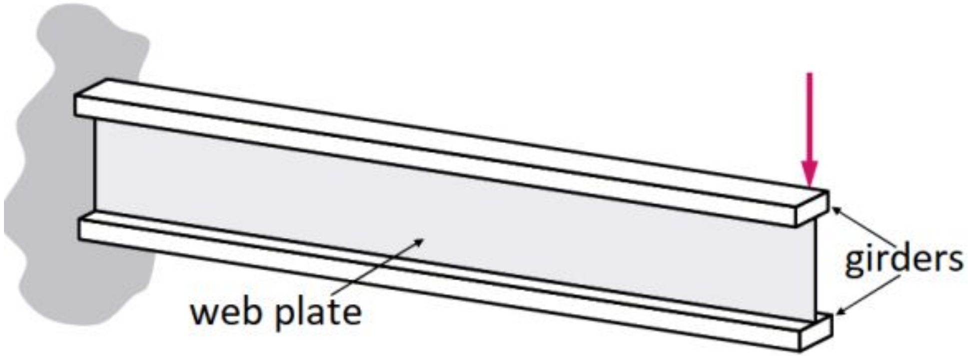
Another design consideration is the distance between the stiffeners. Stiffeners only act locally so you need various stiffeners to make sure the whole sheet is stable





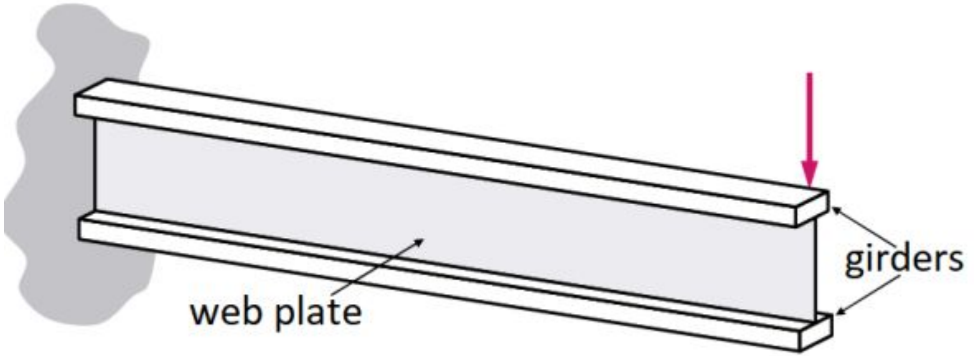
Beam

With the introduction of sheet material, we can use more efficient structures than trusses. We can separate which parts take up the normal forces (tension and compression) and which take up shear loads
We use **beams**

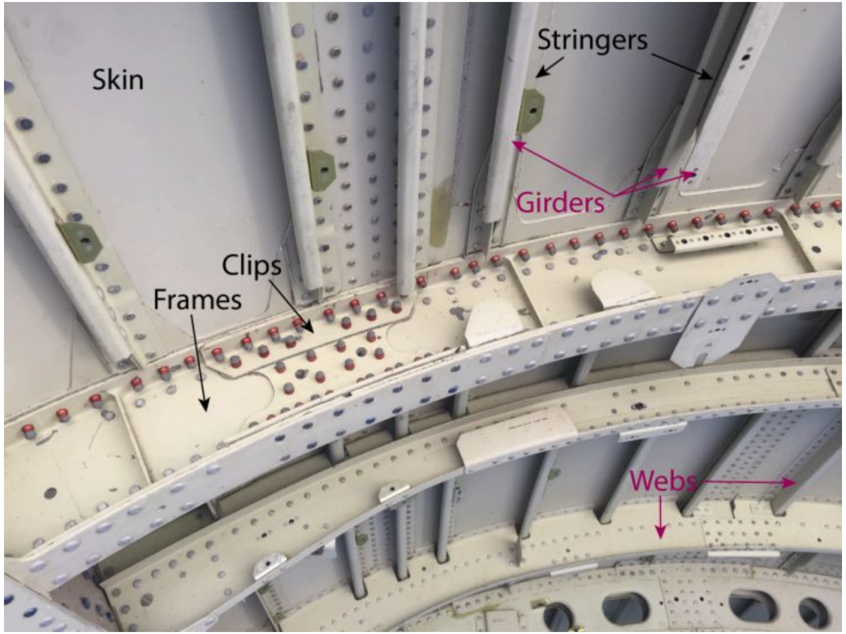


Beam

The (bending) loads are taken up by the girders, and the sheet performing the diagonal function is called the web plate

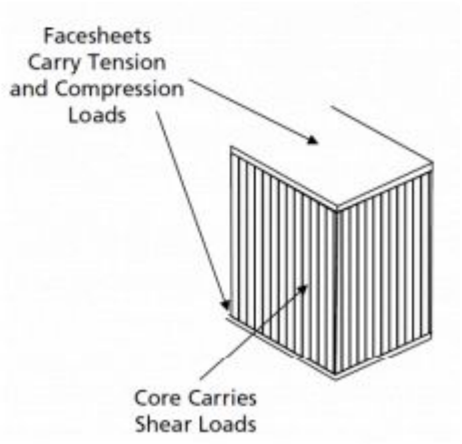


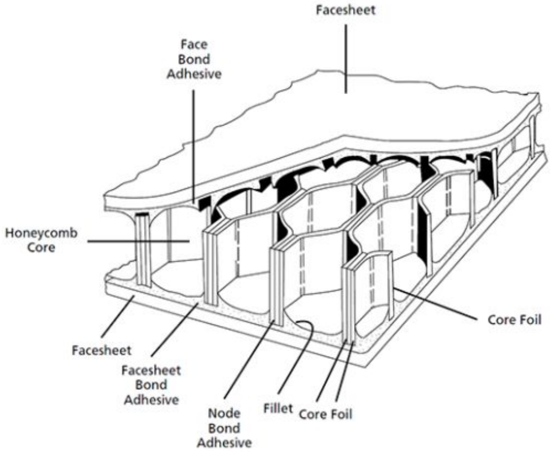
Beams in aviation



Sandwich structure

The sandwich structure contains a low density core in-between two face sheets.
The sheets carry normal loads, and the core carries shear loads

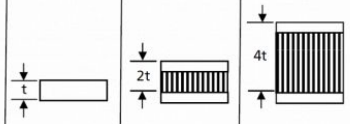




Sandwich structure

- **Advantages:**
  - Inherent high bending stiffness (Used as floors in aircraft)
  - Low weight increment

- **Drawbacks:**
  - Complex and expensive (for example complex joints)
  - Low durability (moisture absorption)



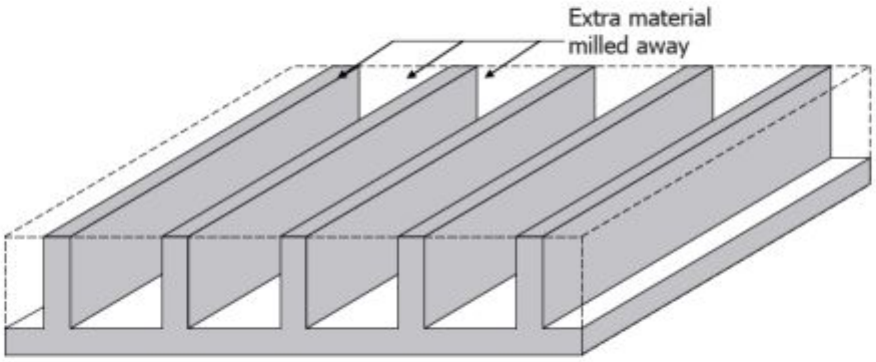
	Solid Material	Sandwich Construction	Thicker Sandwich
Stiffness	1.0	7.0	37.0
Flexural Strength	1.0	3.5	9.2
Weight	1.0	1.03	1.06

Integrally stiffened structures

You can add stiffeners to a shell structure by riveting **or**

Integrally stiffened structures

You can add stiffeners to a shell structure by riveting/ bonding **or**
taking a thick plate and machining away the material between stiffeners



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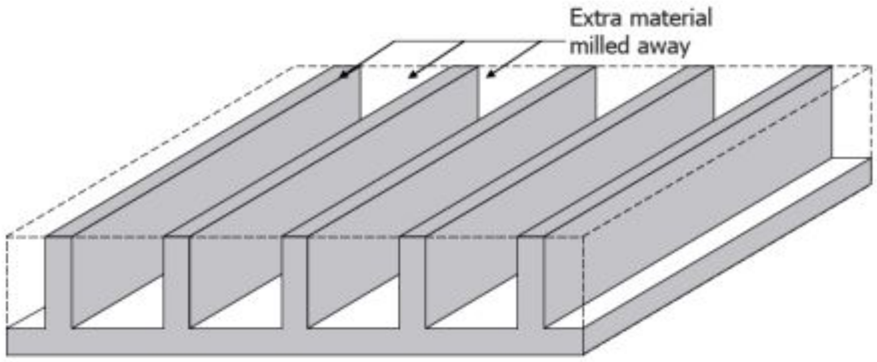
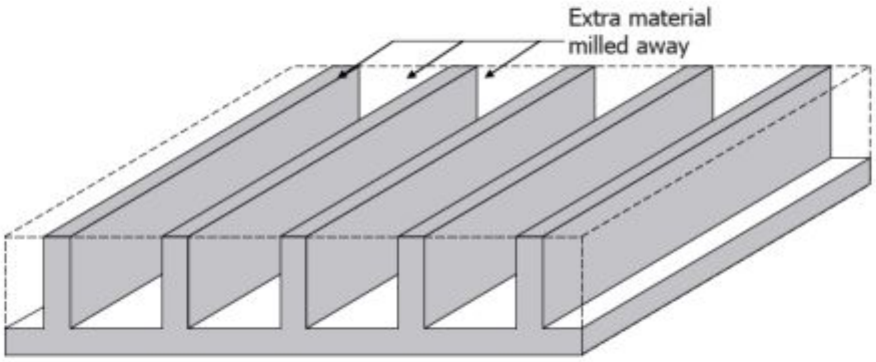


this is called an integrally stiffened structure

Integrally stiffened structures

More cost-effective for thick reinforced plates (so used in large aircraft)

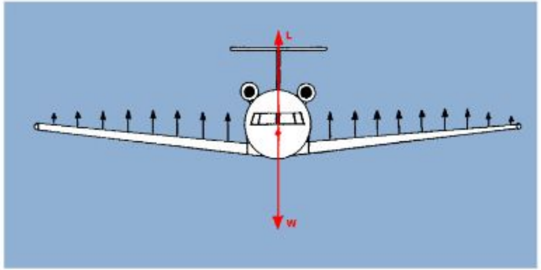


But less damage-tolerant (as cracks spread out throughout whole component)

Fuselage structures

Functions of a structure:

Carry loads



Protect passengers and cargo

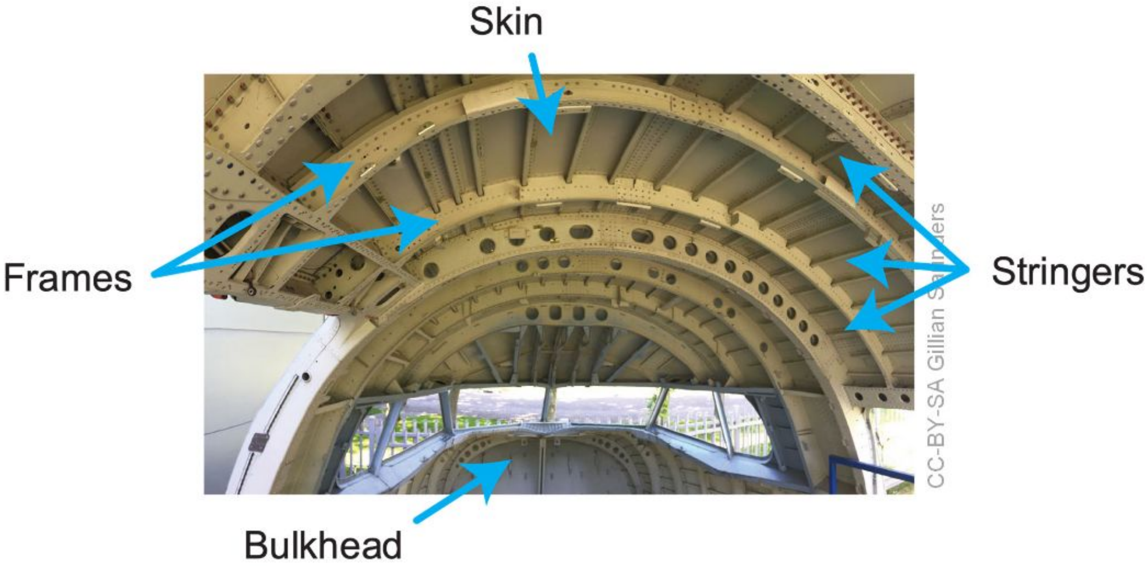
Maintain aerodynamic shape

Connect subsystems

Fuselage structures

The fuselage structure is in practice a thin-walled pressure vessel.
A cylindrical stiffened shell
The main elements are:

- Fuselage skin
- Frames
- Longerons
- Stringers
- Bulkheads
- Splices and joints



Fuselage structures

The fuselage skin protects the payload inside the aircraft and keeps the aircraft pressurized, withstanding shear loads. It also provides some longitudinal stiffness.

Bulkheads are needed to make the vessel air-tight

Splices and joints are used to join different parts together

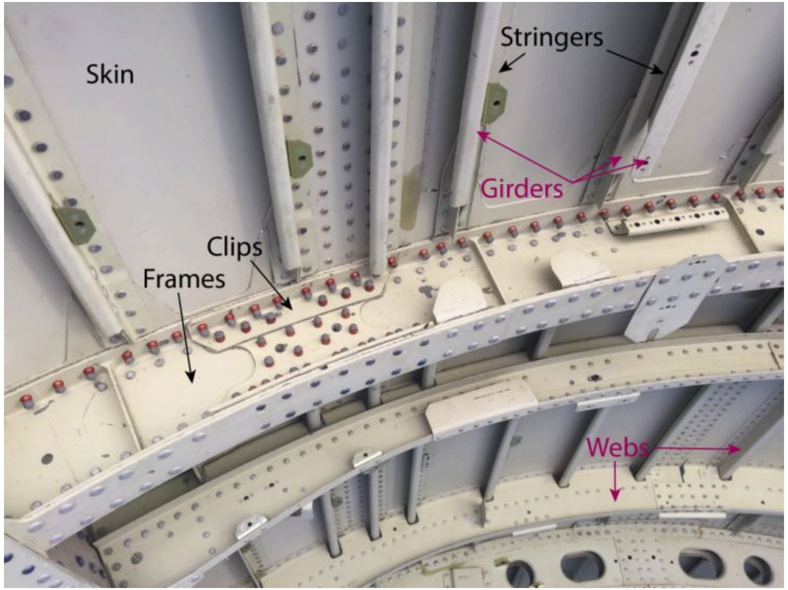
Exercises

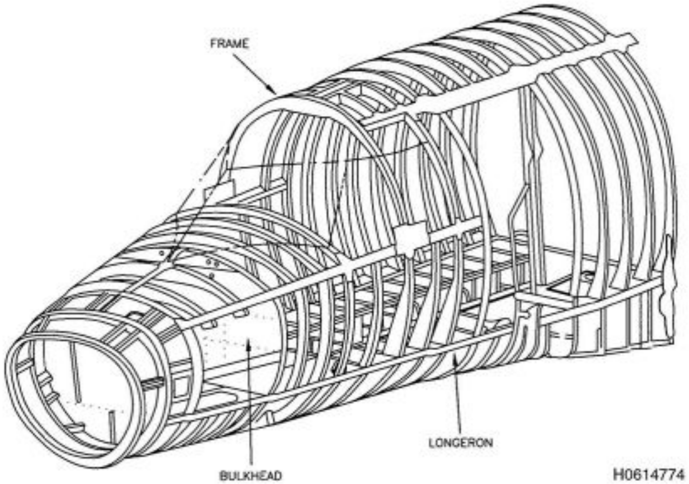
7. You are part of a team tasked with designing a hydrogen powered aircraft and you are currently focused on the design of the pressure tanks needed to carry the hydrogen fuel at a pressure of 300 bar. The initial design is a thin-walled spherical pressure tank, but your colleague is suggesting that the tank design should be changed to a thin-walled cube to maximize the volume of the tank. Do you agree with the idea of your colleague? Why or why not?

Fuselage structures

Stringers provide longitudinal and bending stiffness
They are small stiffeners
Aircraft have large stringers in charge of carrying large loads, with stringers between them

Large stringers in longitudinal direction are called longerons
Large stringers in cylindrical direction are called frames

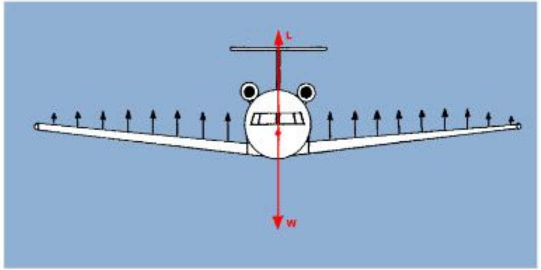

Skin
Stringers
Girders
Clips
Frames
Webs


FRAME
LONGERON
BULKHEAD
H0614774

Wing structures

Functions of a structure:

Carry loads



Protect passengers and cargo

Maintain aerodynamic shape

Connect subsystems

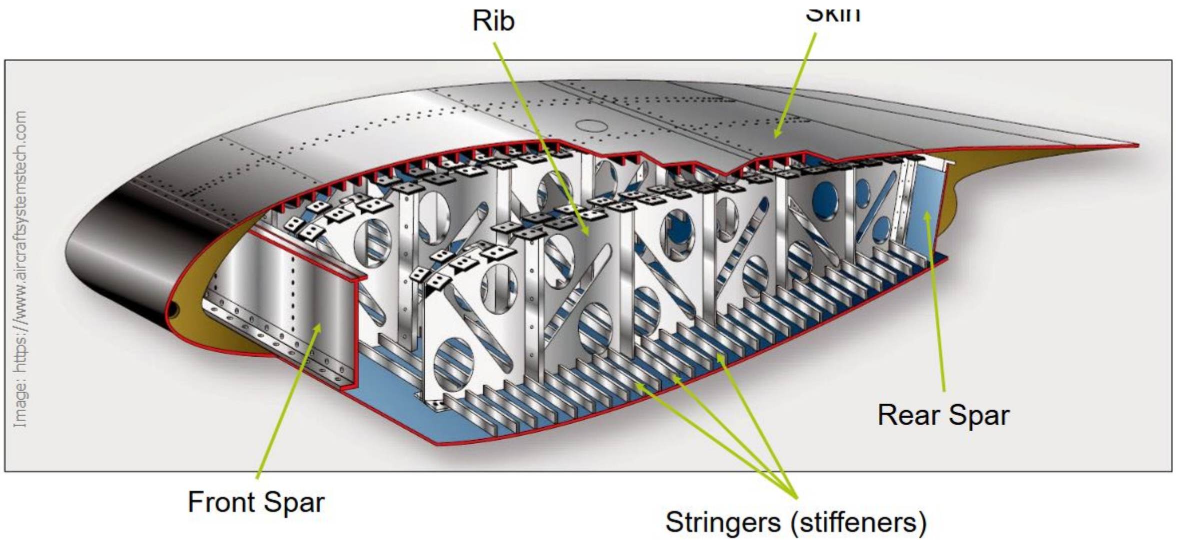
Wing structures

Wing also use the stiffened shell concept, but due to the different shape its parts have different names:

- Wing skin panels
- Ribs
- Spars
- Stringers

Wing skin panels are like the fuselage skin: they carry some shear and protect the inside of the wing
Ribs are like frames
Spars are like longerons

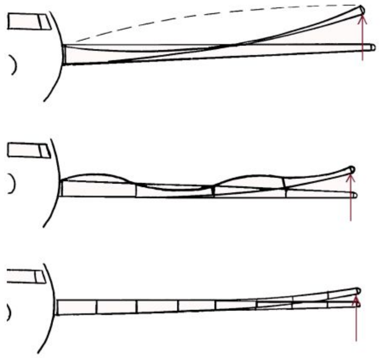
Wing structures



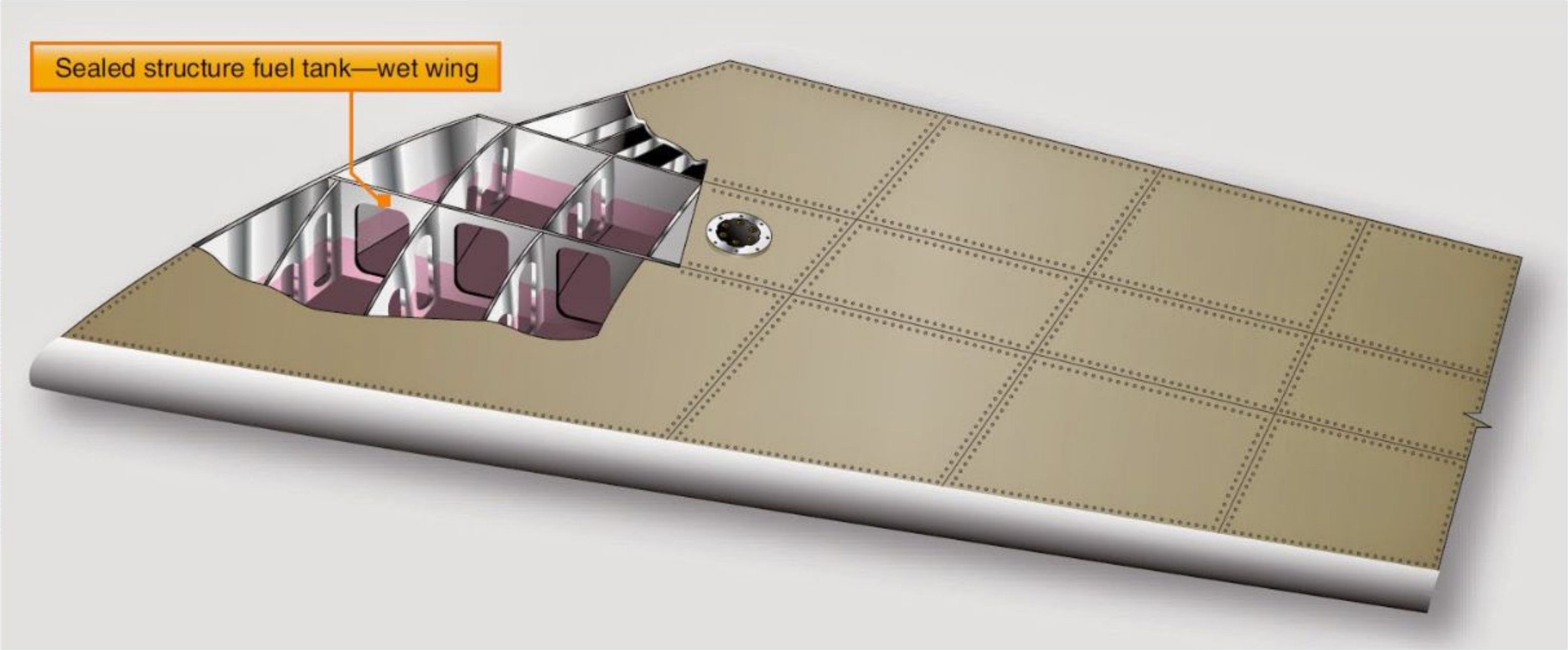
Ribs

Ribs have several functions:

- Maintain aerodynamic profile of wing
- Transfer the aerodynamic and fuel (vertical) loads acting on skin to the rest of the wing structure
- Stability against panel crushing/ buckling
  - Brazier effect - the cross-section tending to ovalise under bending
- Introduce local loads (such as landing gear)
- Sectioning fuel tanks



Wet Wings



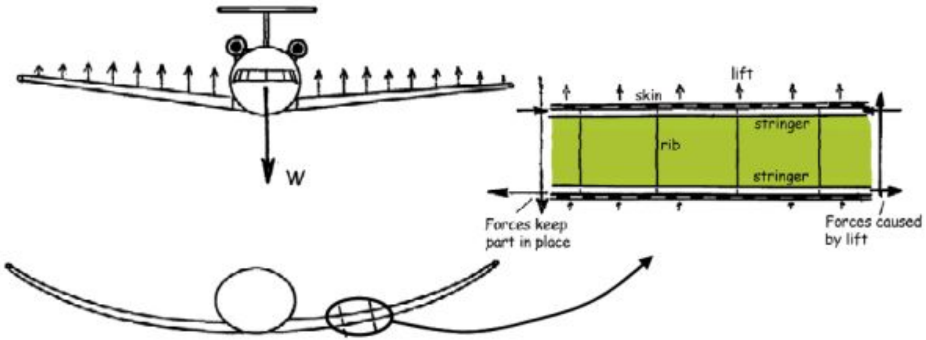
Ribs

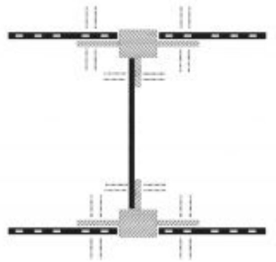
The rib type, material, design and spacing will depend on:

- Loads
- Design philosophy
- Available equipment and experience
- Costs

Spars

The main function of spars is to carry the wing bending loads





The typical shape of spars are I-beams: the sheet carries shear loads and the girders carry normal loads

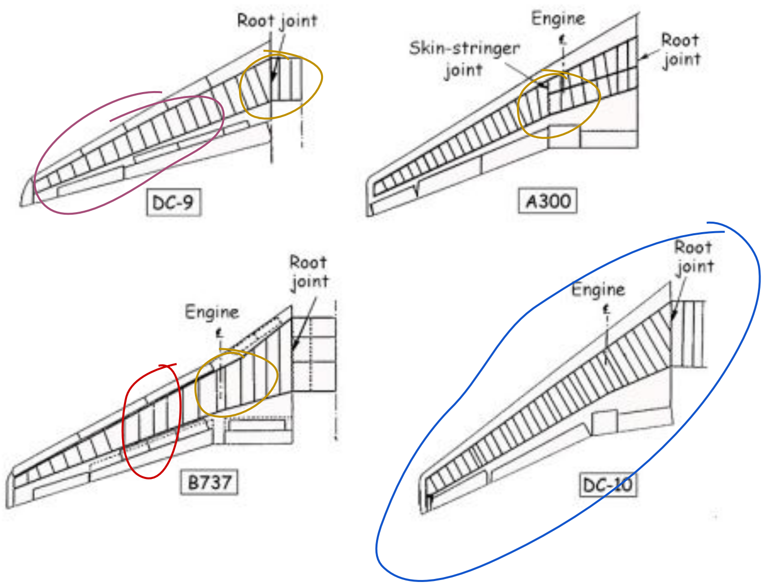
Spars

The loads near the root will be much larger than in the tip
This is because the spar has to withstand the bending moment of the **whole wing**

So usually the root will have thicker spars, or even an additional spar

Variations

Some spars are straight all along the wing, other wings have “kinks”
Some ribs are placed flight direction,
(to keep aerodynamic profile)
others are placed perpendicular to the spar
(because they will be shorter)



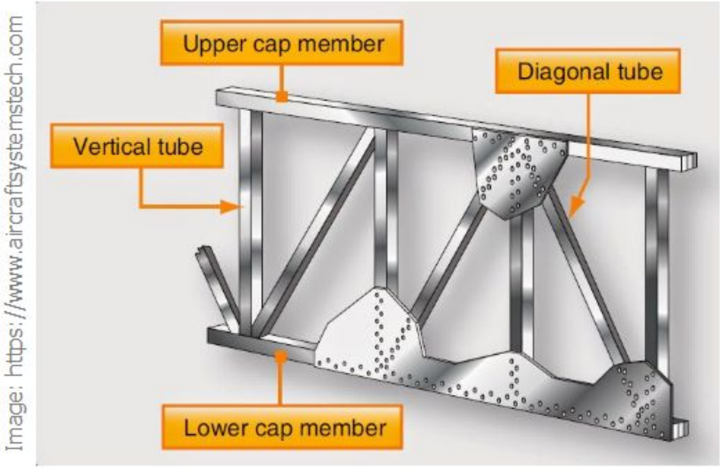
Spars

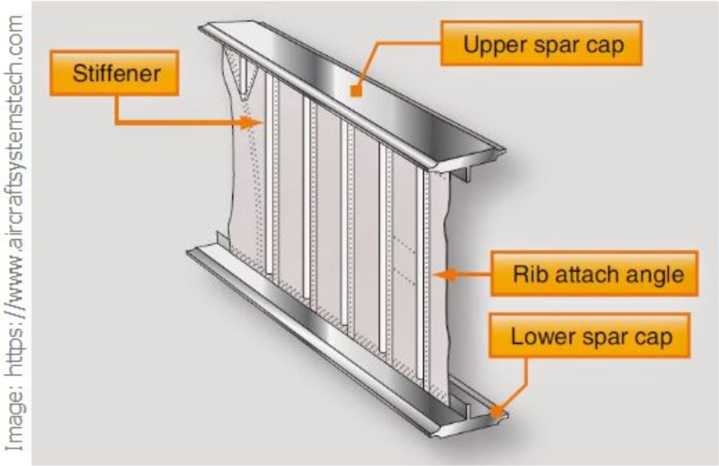
As with ribs, the spar type, material, design and spacing will depend on:

- Loads
- Design philosophy
- Available equipment and experience
- Costs

Variations

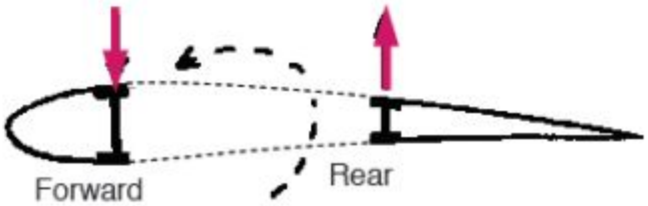
There exist many variations to wing structures.
Spars can be a truss structure or a stiffened shell structure





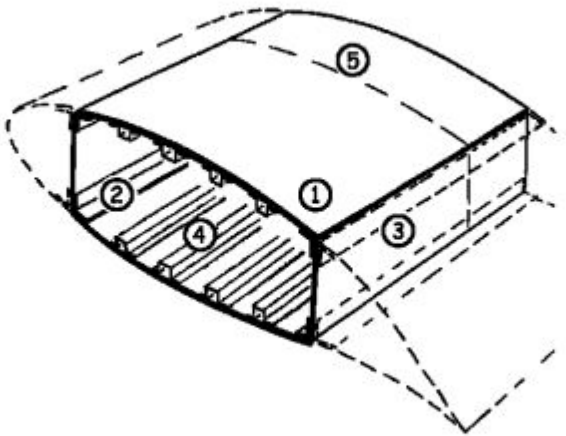
Spars

All aircraft will usually have 2 spars (front and rear) to carry torsional moments

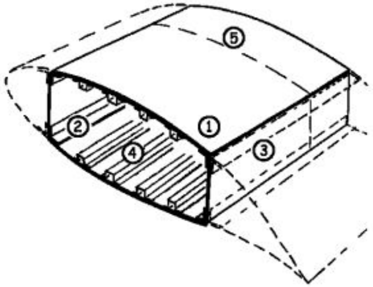


Torsional box

However torsional moments can be carried more efficiently by **closed** torsional boxes



Torsional box



A torsional box is preferred over 2 spars because:

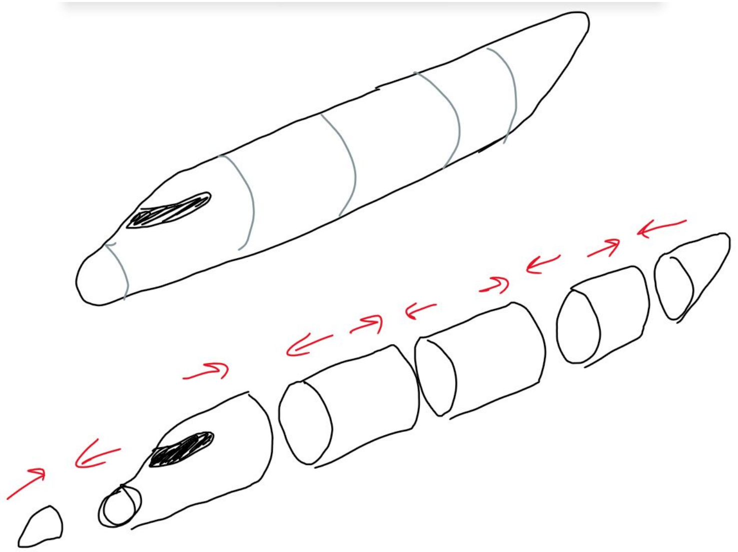
- No required supports/ struts
- At a given wing thickness, the wing can be longer
- You can engineer torsional moment and bending moment separately

All in all, torsional boxes are more lightweight in comparison with the 2 spar concept

Structural Details

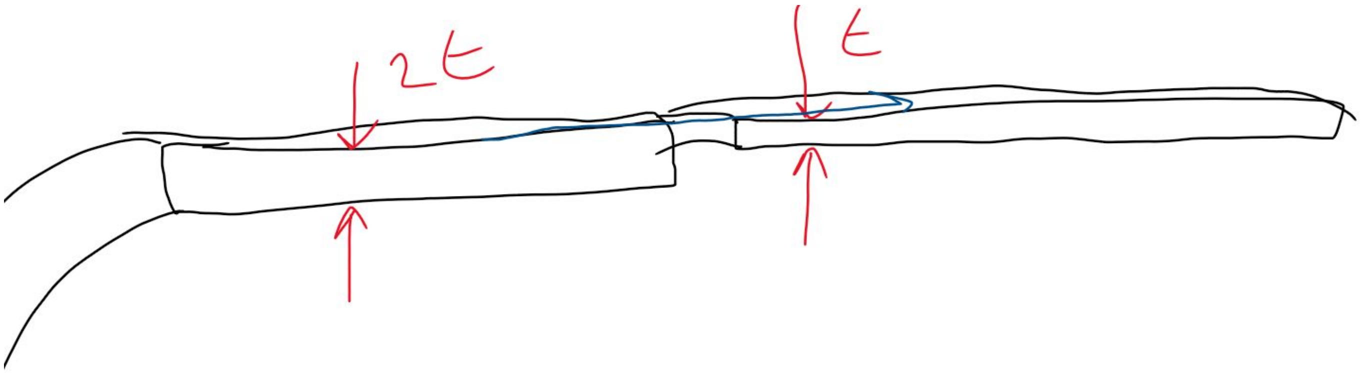
Stringer joggling

The fuselage can be seen as various barrels put together in a vessel

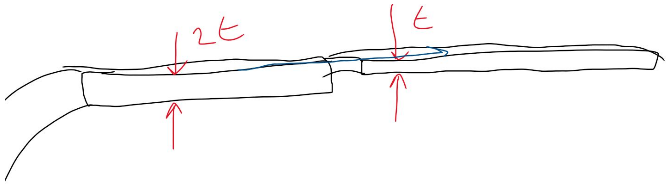


Stringer joggling

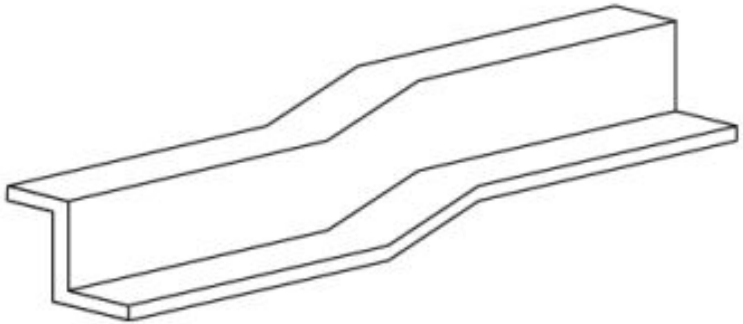
Not all barrels will have the same thickness, and because the outside of the fuselage has to be continuous (due to aerodynamics), there will be a step inside



Stringer joggling



This can be solved using stringer joggling as shown:



Joggling of a stringer implies a shear deformation of the stringer

Stringer coupling

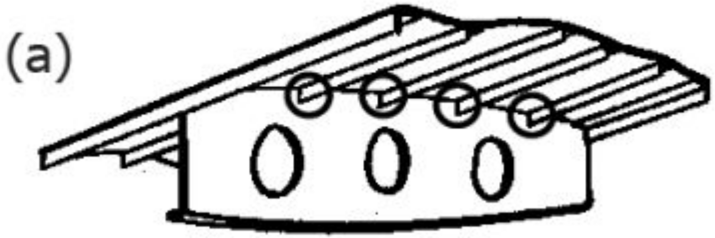
I don't really know how to explain this so let's look at the book

pg 111

Book

Stringer frame and stringer rib intersections

When a string intersects with a rib or a frame, you either cut the stringer (so the rib/ frame is uninterrupted),



Stringer frame and stringer rib intersections

When a string intersects with a rib or a frame, you either cut the stringer (so the rib/ frame is uninterrupted),

you cut the frame/rib so that the stringer is uninterrupted,

Stringer frame and stringer rib intersections

When a string intersects with a rib or a frame, you either cut the stringer (so the rib/ frame is uninterrupted),

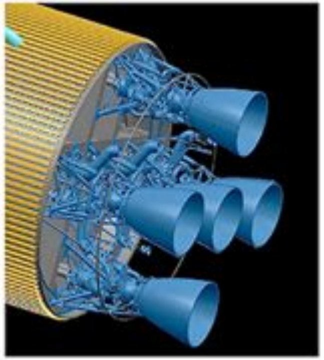
you cut the frame/rib so that the stringer is uninterrupted,

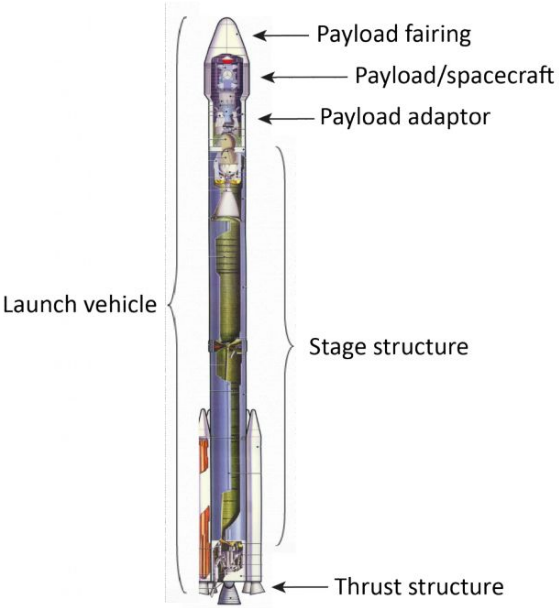
or you don't cut any, and elevate the frame/rib to let the stringers pass through uninterrupted

Launch vehicle structures

Thrust structures

The high thrust loads are introduced very concentrated and have to be distributed into the stage via a conical structure



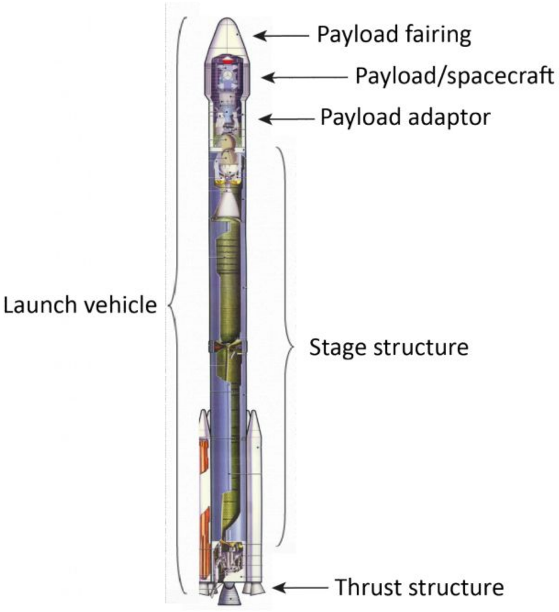


Stage structures

Rocket launchers consist on multiple stages, each with a rocket propulsion system

When the fuel runs out, the stage is disposed and the next stage kicks in

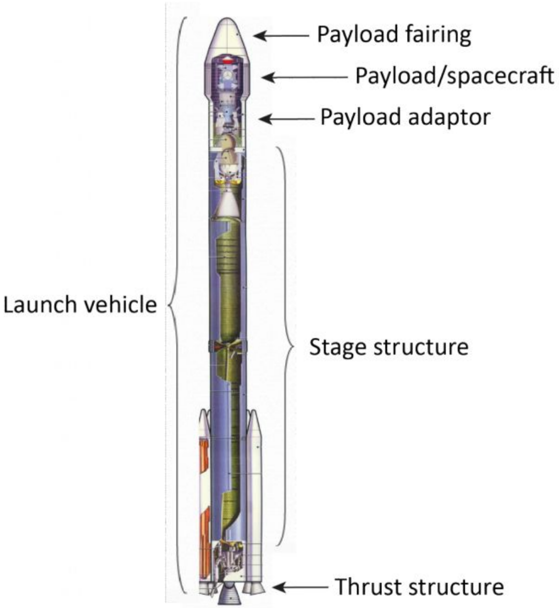
They can be place in series (main stages) or in parallel (booster stages)



Adaptors

Transfers the loads from the stages to the payload

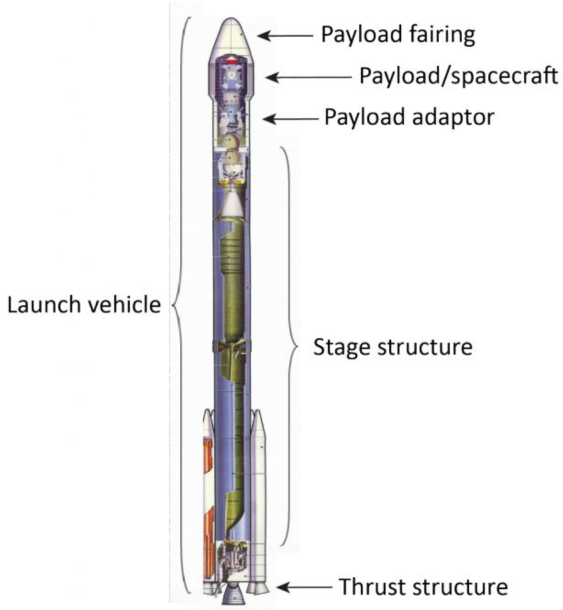
Similar conical shapes to the thrust structures are used



Payload fairing

Protects the payload during launch and ascent against the aggressive environment (aerodynamic heating and pressure)

Once you get out of the atmosphere, the payload is disposed to decrease mass



Spacecraft structures

Spacecraft structures

Although spacecraft operate in space, the main design considerations have to do with the launch phase - when the loads are maximum

The thrust force provided at the bottom of the spacecraft has to be transferred to the structure

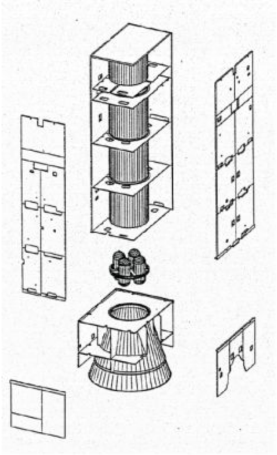
2 structures are used:

- Central cylindrical shell structures
- Strutted structures

Central cylindrical shell structure

All load provided by thrust and inserted via the adaptor is transferred through a central cylinder that is primarily dimensioned for this load case.

All systems within the spacecraft are attached at strong points directly to the cylinder

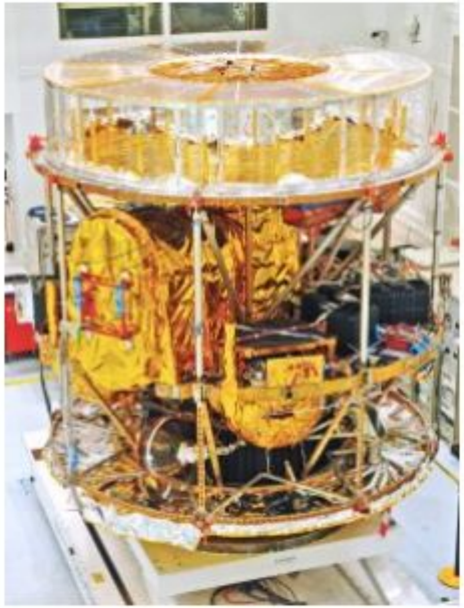


Strutted structures

Truss structures insert the thrust loads using struts (beams)

The trusses should not bulge out in compression, but remain stable to be able to transfer the load to the remaining structure.

This is the primary function of the struts so they can be optimized for this function



Exercises

1. Between 1903 and 1940, a significant change has taken place on the skin of aircraft wings and fuselages. What change is this?

   a) The wing and fuselage skin have become increasingly thinner
   b) The wing and fuselage skin went from non-loaded to load-carrying
   c) The fuselage skin has been optimised to incorporate windows
   d) The wing and fuselage skin were eventually made of different materials

Exercises

2. Indicate whether the following statements are True (T) or False (F):

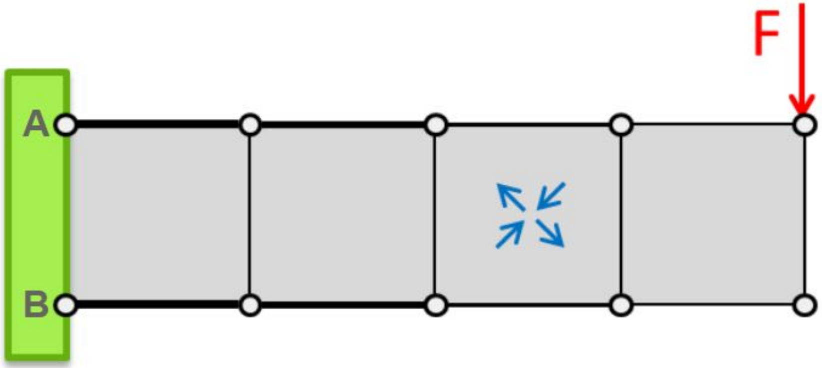
   - The primary function of a **wing rib** is to increase the torsional stiffness of the wing
   - The primary function of a **spar web** is to carry transverse shear loads in the wing
   - Buckling of load carrying skins (fuselage and wing skins) **is not** a catastrophic failure mode for aircraft

Exercises

3. What does the abbreviation PSE stand for when we talk about aircraft structures?

Exercises

6. Indicate whether the following statements are True (T) or False (F) for the web-stiffened truss structure shown below:



- The bottom side of the web-stiffened truss is in tension
- The normal force along the top side of the web-stiffened truss is constant
- The shear force in the web along the length of the web-stiffened truss is constant
- In the truss shown above, the directions of the web forces shown are correct